

DELIMITER \$\$

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CREATE TRIGGER trg_after_update_enrollment
AFTER UPDATE ON enrollment
FOR EACH ROW
BEGIN
    DECLARE total_points DECIMAL(10,2) DEFAULT 0;
    DECLARE total_credits INT DEFAULT 0;
    DECLARE calculated_gpa DECIMAL(5,2);
    SELECT
        IFNULL(SUM(g.grade_point * c.credits), 0),
        IFNULL(SUM(c.credits), 0)
    INTO total_points, total_credits
    FROM enrollment e
    JOIN grade g ON e.grade_letter = g.grade_letter
    JOIN course c ON e.course_id = c.course_id
    WHERE e.student_id = NEW.student_id;
    IF total_credits > 0 THEN
        SET calculated_gpa = total_points / total_credits;
    ELSE
        SET calculated_gpa = 0;
    END IF;
    INSERT INTO gpa_record (
        student_id,
        semester,
        academic_year,
        semester_gpa,
        cumulative_gpa,
        total_credits_earned
    )
    VALUES (
        NEW.student_id,
        NEW.semster,
        NEW.academic_year,
        calculated_gpa,
        calculated_gpa,
        total_credits
    );
END $$
```

DELIMITER ;

short explanation.

This trigger is executed automatically after a new record is inserted into the enrollment table. It recalculates the student's cumulative GPA by joining the enrollment, grade, and course tables to compute total grade points and credits. The updated GPA is then stored in the gpa_record table. This ensures that GPA records remain up-to-date without requiring manual recalculation.